Institute for Computer Science VI, Autonomous Intelligent Systems, University of Bonn

Dr. N. Goerke
Friedrich-Hirzebruch-Allee 8, 53115 Bonn, Tel: +49 228 73-4167
E-Mail: goerke (at) ais.uni-bonn.de
http://www.ais.uni-bonn.de/WS/4204_L_NN.html

Exercises for module Technical Neural Networks (MA-INF 4204), WS26/27

Assignments Sheet 3, due: Tuesday 3.11.2026

27.10.2026

Assignment 12 (4 Points)

Derive a new learning rule * for a Multi-Layer-Perceptron.
Start from the new objective function (cost function, error function) E^* and derive the new learning rule in analogy to Backpropagation of Error. Write down all calculation steps, and explicitly write down the formulas for calculating the δ^* in output- and hidden layer.

$$ {}^pE^*(w_{ij}) = \sum_{m=0}^{M} \left({}^p\hat{y}_m \; - \; {}^py_m \right)^4 $$

Assignment 13 (3 Points)

Apply the learning rule Backpropagation of Error to calculate the weight change ΔW_{nh} for a weight w_{nh} at a hidden layer neuron h in a 3 layer N-H-1MLP. The transfer function in hidden and output layer is the hyperbolic tangent (tanh). The learning rate is $\eta = 0.1$, the teacher value is $\hat{y}_{m=1} = 0.2$, the output value is $y_{m=1} = 0.8$, the weight at the output neuron coming from hidden neuron h is $w_{hm} = 2.5$, the output of the hidden neuron h is $o_h = 0.3$, the input to the hidden neuron is $x_n = 15$.
Write down all intermediate steps of the calculation.

Assignment 14 (3 Points)

A simple 2-1 neural network is given with 2 inputs x_1, x_2, one output neuron with hyperbolic tangent, and the three weights (w_0, w_1, w_2). The objective for the network is to implement the Boolean function XOR, the input values are 0.0 or 1.0, the desired output values $\hat{y}$ are -1 or $+1$.

Depict the (global) error surface $G(w_0, w_1, w_2)$, for the BIAS-weight fixed to $w_0 = 2.0$, and the weights $-10.0 \leq w_1, w_2 \leq +10.0$.

Extra task (no extra points):
Change the setup to desired values 0 and 1, and the transfer function to be *relu*.

Assignment 15 (2 Points)

Assignment 15 has to be operated using AI help.

Report which AI-System helped you, which prompts you have decided to use. What the percentage did the AI-system contribute to your solution? In case you had to re-set your AI-session to get an answer, report why, when and how often. Have you discarded some of the AI-generated answers? If yes, Why?

Compare the two Backpropagation of error variants:
single step learning and *cumulative learning*.
Describe both and discuss advantages and disadvantages of the two methods.